# bandit

September 14, 2026

[2]:
```
import numpy as np
import matplotlib.pyplot as plt


class Bandit:
    def __init__(self, arms=10):  # arms =
        self.rates = np.random.rand(arms)  #          ( )

    def play(self, arm):
        rate = self.rates[arm]
        if rate > np.random.rand():
            return 1

        else:
            return 0


class Agent:
    def __init__(self, epsilon, action_size=10):
        self.epsilon = epsilon  #        (   )
        self.Qs = np.zeros(action_size)
        self.ns = np.zeros(action_size)

    #
    def update(self, action, reward):
        self.ns[action] += 1
        self.Qs[action] += (reward - self.Qs[action]) / self.ns[action]

    #    ( -   )
    def get_action(self):
        if np.random.rand() < self.epsilon:
            return np.random.randint(0, len(self.Qs))  #
        return np.argmax(self.Qs)  #


if __name__ == '__main__':
    steps = 1000
```

```
    epsilon = 0.1

    bandit = Bandit()
    agent = Agent(epsilon)
    total_reward = 0
    total_rewards = []  #
    rates = []          #

    for step in range(steps):
        action = agent.get_action()   #
        reward = bandit.play(action)  #
        agent.update(action, reward)  #
        total_reward += reward

        total_rewards.append(total_reward)       #
        rates.append(total_reward / (step + 1))  #

    print(total_reward)

    # [  1-12]
    plt.ylabel('Total reward')
    plt.xlabel('Steps')
    plt.plot(total_rewards)
    plt.show()

    # [  1-13]
    plt.ylabel('Rates')
    plt.xlabel('Steps')
    plt.plot(rates)
    plt.show()
```

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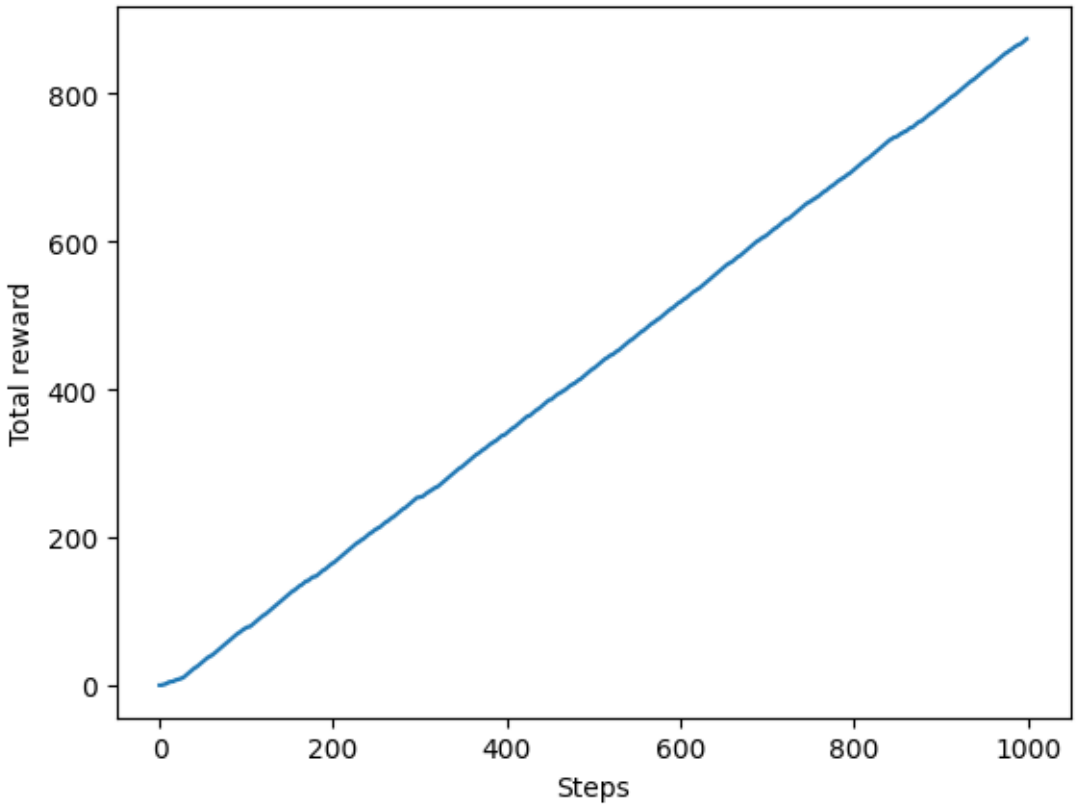
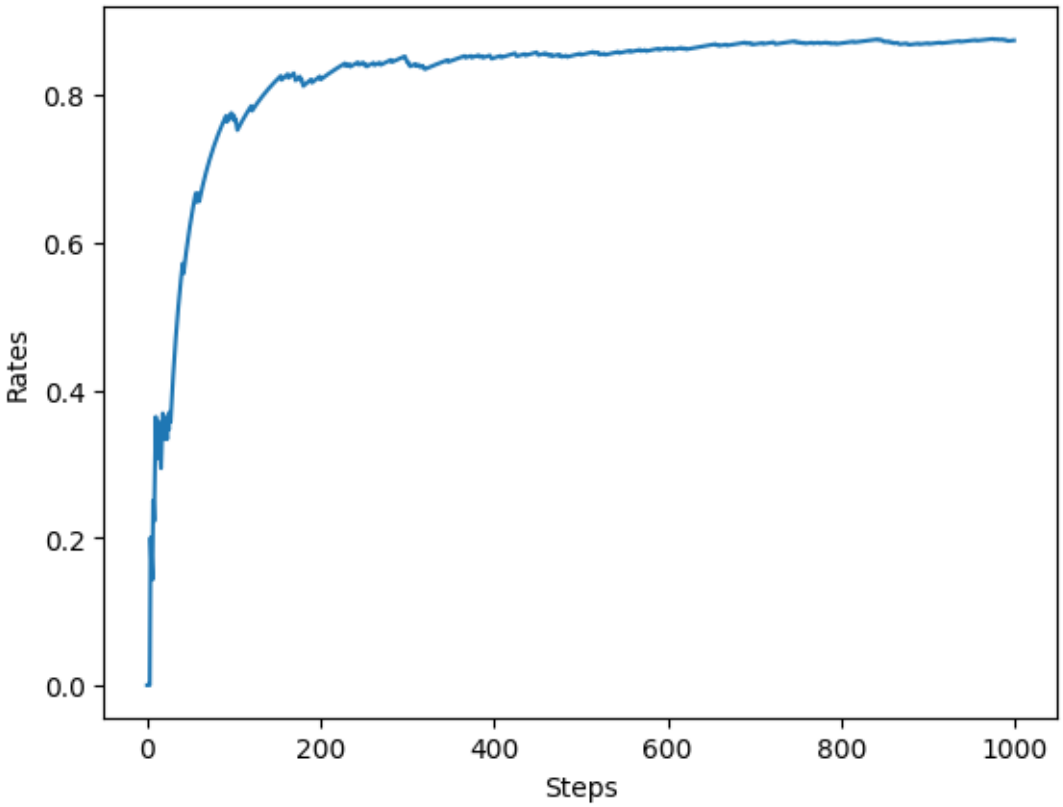

Total reward
0
200
400
600
800
Steps
0
200
400
600
800
1000


0.8
0.6
0.4
0.2
0.0
Rates
0
200
400
600
800
1000
Steps